Early Autism Screening in Children Using Facial Recognition

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ABSTRACT

Facial data of children aged 2 to 10 years was collected from three institutions in a large city; an autism rehabilitation center, a rehabilitation hospital, and an inclusive kindergarten. The dataset comprised facial data of 65 children diagnosed with autism, and 47 children with typical developmental. We employed the VGG-16 model to develop a facial feature recognition-based early screening system, which involves feature extraction and image processing of eyes, eyebrows, noses, and mouths. The data was processed and categorized using a Convolutional Neural Network (CNN) model, and the accuracy of this algorithm was validated. Cross-testing with the public database Kaggle and our dataset demonstrated an accuracy rate of up to 94% for the current training set. This indicates that the model trained by our system is proficient in classifying children's facial data and maintains high precision on our database.

CCS CONCEPTS

• **Human-centered computing → Accessibility systems and tools**.

KEYWORDS

Children with autism, Assistive technology, Screening and earlier diagnosis, Facial recognition, Algorithm

ACM Reference Format:
Qin Wu, Xiangming Xiao, Yugui Liu, Mark Billinghurst, and Suranga Nanayakkara. 2024. Early Autism Screening in Children Using Facial Recognition. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24), May 11–16, 2024, Honolulu, HI, USA.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3613905.3651045

CHI EA '24, May 11–16, 2024, Honolulu, HI, USA

ACM ISBN 979-8-4007-0331-7/24/05
https://doi.org/10.1145/3613905.3651045

1 INTRODUCTION

Recent data reveals that the prevalence of autism in children has reached 1 in 36 [7]. Early diagnosis of autism is crucial as it leads to early intervention, significantly improving the prognosis for children with ASD [20]. However, traditional autism assessment scales, such as behavioural observation and parental reporting [8, 17, 23, 25], are often inefficient and limited by factors like delayed detection, particularly in children with less obvious symptoms or from diverse cultural backgrounds [15]. Consequently, the field of autism diagnosis remains fraught with unresolved issues, highlighting the need for more efficient, accurate, and universally applicable diagnostic tools. Autism detection is also important for enabling effective human computer interaction between children and technology, such as face scanning and emotion perception analysis [26], Eye Tracking Data Analysis [5], pre-speech vocalizations [4]. Artificial Intelligence and Machine Learning technologies [2, 14, 24] have shown potential in the early screening process for children with autism, offering possibilities for detecting ASD characteristics such as subtle facial expressions and emotional responses [3, 13]. These can be challenging for clinicians and therapists to identify consistently.

This study aims to demonstrate the potential of facial recognition technology based on the VGG-16 model for early screening of children with autism. The primary work of this project involved collecting and establishing a database, training data, and sample testing. We completed facial data collection at three institutions in Beijing, China; an autism rehabilitation centre, an autism hospital, and an inclusive kindergarten, involving 65 children with ASD and 47 typically developing (TD) children. The children's ages ranged from 2 to 10 years, with an average age of 4.57 years. The ASD dataset comprised 55 males and 10 females, while the TD dataset included 34 males and 13 females. Based on the paediatrician' CARS-2 [23] diagnostic results and assessment diaries, we selected 5,000 facial images each from children with ASD and typically developing children for this study. We designed and developed a facial recognition autism early screening system based on the VGG-16 convolutional neural network. The system includes creating algorithms, collecting and processing facial data, screening tests, and windows application. We used Histogram Equalization, Noise Reduction, and Normalization for image preprocessing. The VGG-16 convolutional neural network model was employed for training the preprocessed image data, followed by model evaluation and cross-validation using functions from the Keras library [10] to ensure the model's accuracy and training loss, ultimately outputting the evaluation results. We conducted cross-validation on both public databases and our own database using various optimizers, achieving a maximum validation accuracy of 94%. This underscores the positive impact of our system's modeling approach in the screening of children with autism. This database significantly enhances the identification of facial features in East Asian children with autism, providing a potentially fast and effective method for early detection and screening. This study's contribution lies in its potential to provide a non-invasive, efficient, and universally accessible tool for early autism screening, thereby enriching the technological practices in current ASD diagnostic practices.

2 RELATED WORK

Traditional methods for autism diagnosis rely heavily on behavioral observation and parental reports, using tools such as the Modified Checklist for Autism in Toddlers (M-CHAT-R/F) [21], Verbal Behavior Milestones Assessment and Placement Program (VB-MAPP) [25], the Parents' Evaluation of Developmental Status [8], Autism Diagnostic Observation Schedule [17], and the Childhood Autism Rating Scale [23]. While widely used, they face limitations in early detection, particularly in children under two and those with subtle symptoms or from diverse cultural backgrounds [29]. Additionally, the scarcity of medical professionals specialized in autism diagnosis contributes to late detection in many cases, missing the crucial early intervention window [18]. Emerging research has identified facial expressions and movements in children with autism as potential early indicators of the disorder [1, 9]. For example, children showing atypical facial expressions that mimic expressions of happiness, sadness, and fear could provide potential cognitive markers for quantifying syndrome manifestations in children with autism [13]. iPad-based assessment tools that computationally encode facial movements and expressions can detect differences in emotional expression [3], and facial male features in autistic siblings can help understand the interplay between human physical and cognitive development [28]. These studies provided substantial empirical support, solidified the theoretical foundation of our project, and generated key technical insights necessary to develop our system.

Recent advances in facial recognition technology offer new possibilities for autism screening and assessment [2, 14]. These technologies can detect nuanced facial expressions and emotional responses, which are often overlooked by clinicians. For example, research by Syeda et al. using the Tobii EyeX controller and MATLAB revealed that autistic children focus less on core facial features and struggle with emotion perception, impacting social communication [26]. Wang et al. developed a heterogeneous domain adaptation model for preliminary autism screening, utilizing facial features and behavioral data [30]. Tafasca and colleagues adopted a precision medicine approach, using computer vision and IoT sensing technologies to classify autism subtypes and diagnose ASD based on children's behavior [27]. Lu's team [16] reported an effective method for screening autism in children using facial images, leveraging VGG16 transfer learning on a newly collected clinical dataset, achieving 95% accuracy and a 0.95 F1-score, highlighting the importance of considering racial and ethnic factors for accurate ASD detection. Unlike previous studies, our dataset was assessed and classified by clinicians, ensuring high classification accuracy and feature refinement of a unique dataset of children with autism. Additionally, we cross-validated this dataset with the open-source Kaggle ASD facial image dataset to understand the general ability to detect subtle expressions and emotional responses in people with autism. We recognize that due to racial and regional differences, facial features vary significantly among different autism populations [29]. This can lead to potential inaccuracies when diagnosing autism based on existing public databases such as ASDFace CNN [30] or Kaggle [1],particularly between Asian and Western populations. We address this problem by developing a new database for training and evaluating facial recognition systems. This not only helps improve the diagnostic accuracy of the model in East Asian populations, but also provides valuable data resources for the field of autism research.

3 METHODOLOGY

3.1 Data Set

Data collection took place in Beijing in 2023, with ethical approval obtained from a local university. We established contact with parents through social media, and all data collection activities were conducted after the parents signed informed consent forms. Participation in the study was voluntary, but each child participant received a blind box toy worth $3 as a gift after completing the facial feature collection. A total of 65 children with ASD and 47 TD children participated from three institutions: an autism rehabilitation centre, an autism hospital, and an inclusive kindergarten completed facial data collection. The children ranged in age from 2 to 10 years, with an average age of 4.57 years. The autistic children dataset included 55 males and 10 females, while the TD group dataset included 34 males and 13 females. Each child with autism had been diagnosed at a local hospital, we also invited a pediatric specialist to reassess the participants using the CARS-2 tool [23] and at the study place. We observed the doctor's assessment process, took notes on the children's typical behaviours, and asked the specialist to provide a medical interpretation of the children's facial features and movement trajectories. Based on these CARS-2 results, we specifically flagged the data of participants with lower scores (children with high needs).

We transported and arranged the study data collection equipment in the rooms where children usually engage in their daily training and learning activities, aiming to minimize the discomfort children might feel in an unfamiliar setting. Before the study began, the child participants were sitting in child-sized chairs facing the monitor. To soothe the children's emotions, we invited experienced therapists to stand beside them for assistance. Once the children were emotionally stable, we initiated the system to display images and began data collection. The children looked at a variety of static images. Based on previous research [6, 19, 22] we selected visual

[1] Autism screening data for toddlers

stimuli images from five categories: fractal patterns, natural landscapes, static objects, social scenes, mixed patterns. There were 25 images with each image displayed for 5 seconds. During the viewing, a 4K high-definition camera positioned above the monitor collected facial data of each child, capturing image sequence frames at a rate of 60 images per second, with each participant's data collection lasting approximately five minutes. Our system ultimately gathered over 200,000 metadata images of facial expressions, including approximately 190,000 from children with ASD and about 11,000 from typically developing children. After data collection, our research team members conducted an initial cleaning and screening of the metadata, removing invalid, duplicate, and interfering data. Based on the diagnoses and assessment logs from specialists, we eventually selected 5,000 facial images each from children with ASD and typically developing children as the dataset for this study.

3.2 System Description and Model Training

Our system comprises a high-definition camera for hardware and a Python-based Windows desktop application for software. It features three main modules: data collection and preprocessing, algorithms and modelling, and screening and evaluation. The data acquisition module employs the Viola-Jones face detection algorithm, utilizing Haar features to differentiate between facial and non-facial areas, integral images for accelerated computation, and the Adaboost algorithm for training a robust classifier. It enhances detection efficiency and employs non-maximum suppression to eliminate overlapping detections. This method effectively detects the feature differences between facial and non-facial areas, thereby quickly and accurately locating facial features in images. For data preprocessing, we employ histogram equalization to adjust the distribution of pixel values and enhance image contrast, noise reduction to decrease the model's sensitivity to noise, and normalization to limit the range of pixel values and alleviate uneven brightness. We utilize the classic CNN architecture, VGG-16, in the algorithm and modelling module, as shown in Figure 1. This model includes 13 convolutional layers and three fully connected layers, comprising five convolutional blocks. Each block consists of consecutive convolutional and pooling layers, followed by three fully connected layers for classification. Each convolutional layer is followed by a ReLU activation function, introducing nonlinearity to the model. After model training is complete in the diagnostic and detection module, the study employs machine learning techniques for model evaluation. We designate one-fifth of the dataset as the validation set and apply cross-validation to assess the model's accuracy and training loss, ultimately providing results indicative of potential ASD or non-ASD in children.

Compared to conventional convolutional neural networks like LeNet [12] and AlexNet [11], the VGG network features a more streamlined architecture, using smaller 3x3 convolutional kernels coupled with 2x2 max pooling layers. This configuration significantly enhances the network's capability to capture finer image details. Additionally, the VGG's simplified and structured architecture facilitates ease in training and optimization and improves its generalization performance. Moreover, the VGG network is characterized by a deeper layering of convolutional stacks, which aids in learning more abstract representations of features, thereby increasing the model's accuracy. SGD and Adam are key optimization algorithms for training neural networks and machine learning models. We tested and found that models trained with Adam exhibit higher accuracy than those with SGD. Adam's advantages include adaptive learning rates and quicker convergence, making it a superior choice for efficient model optimization. In the VGG network construction, Batch Normalization was implemented at the input of each layer to accelerate the training phase, and Dropout regularization was used to avert overfitting scenarios.

The training entails initially inputting preprocessed images into the network's input layer, followed by sequential progression through 13 convolutional layers and five pooling layers. The 3x3 convolutional kernels continuously conduct convolution operations, extracting localized facial features, while the 2x2 pooling windows serve to diminish computational burden while preserving essential features. Post the final pooling stage; image data is flattened, converting multidimensional feature maps into one-dimensional vectors for subsequent processing in the fully connected layers. The process culminates with the image data being directed through two thoroughly combined layers containing 4096 neurons each and a final layer comprising 1000 neurons, employing a softmax activation function for classification predictions, thereby finalizing the model's training. During the testing steps, a subset of 2000 images (1000 each from ASD and non-ASD groups) was randomly selected from the dataset of 10,000 images for validation. Following the training, the model's performance was evaluated using Keras library functions, with the assessment focusing on key metrics such as Training Accuracy, Validation Accuracy, Training Loss, and Validation Loss.

4 RESULTS AND FUTURE WORK

4.1 Testing Results

At this stage, our research has just completed data collection, and the current focus is on data processing and algorithm testing. So this section reports on validation using existing public databases and selects a subset of data from our database for cross-validation. Initially, we conducted feature extraction on children's facial characteristics (including eyes, eyebrows, nose, and mouth) in the database, as depicted in Figure 2. For both ASD and non-ASD children, the raw data underwent a series of preprocessing steps such as denoising, histogram equalization, and normalization to enhance facial features. Subsequently, feature extraction was completed using library functions from OpenCV and dlib. Finally, the extracted feature data were fed into the vgg-16 model for training.

4.1.1 Kaggle Database Testing. For our research, the Kaggle ASD children's facial image dataset of GitHub was selected for first training [2]. This database is dedicated to using computer vision for the detection of autism in children. The Kaggle dataset comprises 4,000 images, evenly split between ASD and non-ASD categories, with 2,000 images each. These images are divided into training and validation sets at a 4:1 ratio, resulting in 3,200 images for training and 800 images for validation. The dataset covers an age range of 2 to 14 years, with a gender ratio of about 3:1 male to female in the ASD group and nearly 1:1 in the non-ASD group. The ratio

[2]Kaggle-Autism

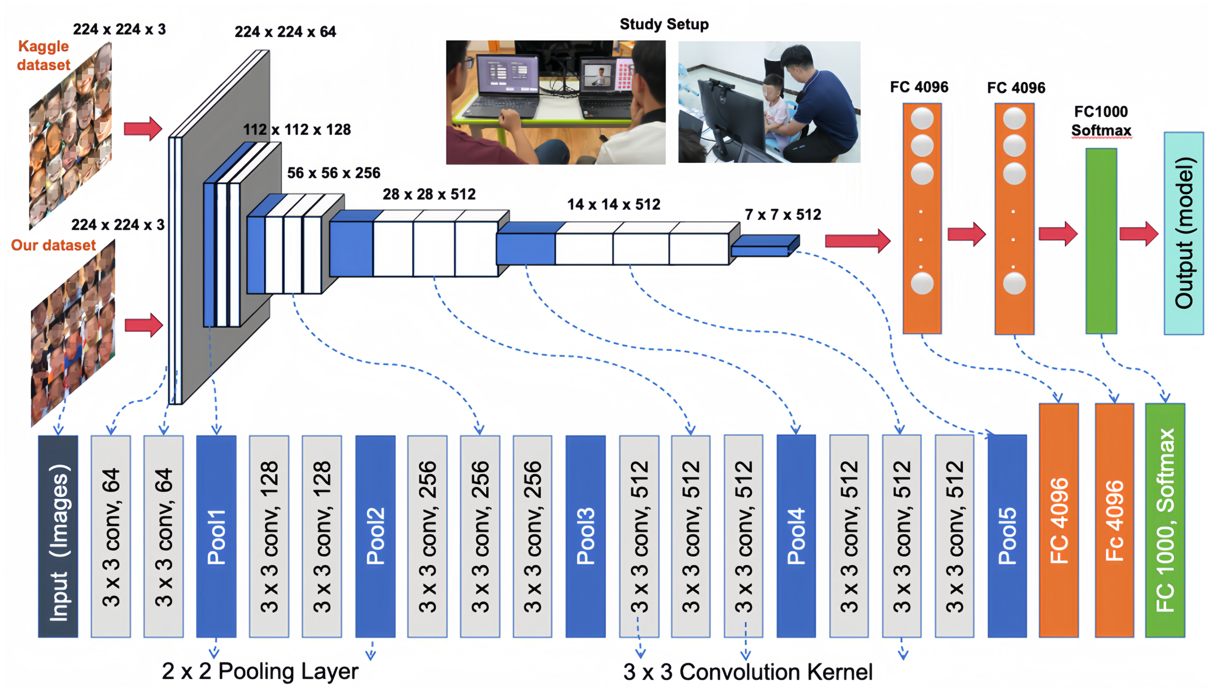

Figure 1: Modelling module VGG-16

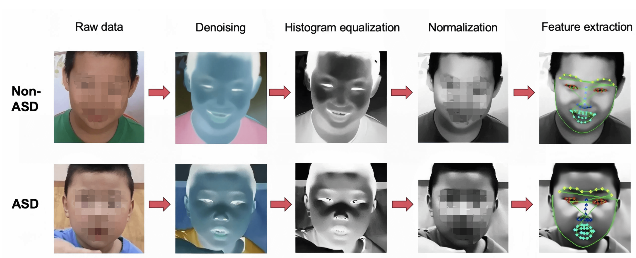

Figure 2: Feature extraction process

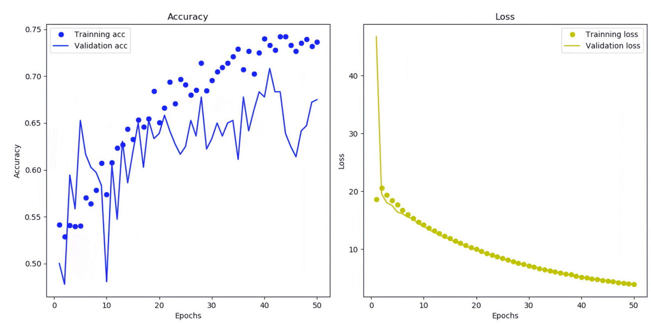

Figure 3: Test result of Kaggle

of Caucasian children to children of other ethnic backgrounds is approximately 10:1. Data preprocessing was conducted prior to training, along with the use of Adam for algorithmic optimization. Throughout the 50-epoch training process, as shown in Figure 3, the peak accuracies for both training and validation sets were observed at the 48th epoch, at 73.5% and 67.5%, respectively. The training set lossed from 18.85 to 4.95, while the validation set loss from 46.78 to 4.93. A notable reduction in loss values was observed after 50 epochs of training, suggesting an enhancement of the model during its training.

4.1.2 Cross-Database Testing. After obtaining the training results from Kaggle, we performed subset selection cross-training with the collected data. We first used the SGD optimizer to enhance the original data algorithmically. Both training and validation set accuracies show some significant fluctuations, but after 50 epochs, these are still relatively low at 54% and 50%, respectively. These low accuracy results indicate that training with unprocessed data significantly impacts model results, preventing effective convergence and reducing accuracy. To solve this problem, we try to use the Adam optimizer to train on the preprocessed data. The study results of this method (shown in Figure 4) show that by the 50th epoch, the accuracy of the training set reaches 92.3%, and the accuracy of the validation set reaches a peak of 94%. The training set lossed from 27.28 to 5.11, while the validation set loss from 42.54 to 5.10. The final results illustrate this approach's variation in training and validation losses. The significant reduction in loss value, coupled with high accuracy, underscores the training effectiveness of the model. This demonstrates the model's proficiency in accurately classifying training data and maintaining robust performance on new validation data, highlighting the success of the applied training protocol. The significant decrease in loss after 50 epochs indicates that the model has improved throughout the training process. The reduction in loss values means that the prediction of the target variable is more accurate and the fit to the training data is improved, further proving that the Adam optimizer is effective in the VGG-16 training process. Table 1 summarizes the experimental results after training with the three schemes, showing that the Adam optimizer preprocessing leads to positive improvements in accuracy and loss values after 50 epochs (Table A1).

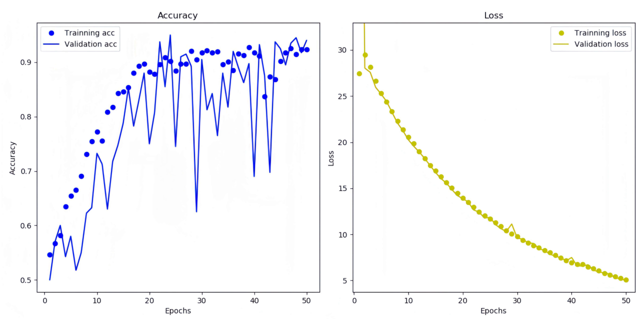

Figure 4: Adam optimizes data results with processed

Table 1: Summary of testing results

Data set	Pre-processing	Optimizer	Training accuracy	Verification accuracy
Kaggle	Yes	Adam	73.5%	67.5%
Our-database	Yes	Adam	92.3%	94%
Our-database	No	SGD	54%	50%

4.2 Discussion

This study demonstrates the potential of machine learning technology to support early autism diagnosis by developing a screening system based on VGG-16 model-enhanced facial recognition technology. Preliminary test results have shown promising high accuracy and sensitivity, especially in analyzing children's facial responses to static images. Using preprocessed data with the VGG-16 deep learning model has resulted in a remarkable 94% accuracy rate, underscoring the system's potential in early, precise autism diagnosis. Moreover, the applicability of this system across diverse ethnic groups highlights its utility in screening East Asian autistic children, thereby addressing a notable gap in the current diagnostic landscape. Despite these advances, the study acknowledges certain limitations. The system's effectiveness is heavily contingent on the quality and diversity of the training data. Insufficient or inadequately preprocessed data could significantly impair the model's generalizability and accuracy. Additionally, the current focus on facial features for autism diagnosis may inadvertently overlook other vital indicators, such as non-verbal behaviour and communication cues. Moreover, this technology's potential applications extend beyond autism. The principles and methodologies developed in this study could be adapted for the early screening and diagnosis of other developmental disorders, where early detection is equally crucial for effective intervention. while this technology marks a significant advancement in the field of autism research, its evolution will be guided by ongoing improvements in machine learning, a more nuanced understanding of autism spectrum disorders, and the integration of multi-modal diagnostic criteria.

4.3 Conclusion and Future work

This project focuses on the early screening technology for children with autism, using facial recognition based on the VGG-16 model. A database was established by collecting facial data from 65 children with ASD and 47 TD children with an average age of 4.57 years. Through data processing and model training, we conducted tests using the SGD optimizer, tests on the Kaggle public database of ethnically diverse children, and tests with the Adam optimizer. Our final validation accuracy reached 94%, demonstrating the potential of our developed data model. In our future work, we plan to explore improvements in facial recognition technology to enhance screening accuracy. This includes optimizing the VGG-16 network model by experimenting with additional convolutional and pooling layers to augment the model's feature extraction capabilities. Furthermore, to address the issue of imbalanced facial data samples between ASD and TD children, we will employ oversampling and undersampling techniques to balance the data distribution. We also aim to recruit more participants for data testing to validate the accuracy and effectiveness of our system. We hope to categorize different types of autistic children, such as those with high needs, low needs, and typical autism, thereby enabling more detailed classification and intervention strategies. This technology holds promise for promoting early intervention strategies, crucial for improving long-term outcomes in autistic children. Beyond autism, similar technology might be applicable for early screening and diagnosis of other developmental disorders.

ACKNOWLEDGMENTS

We would like to thank all individuals who took part in our study. We also thank Zian Xu, Rong Zhang and Yan Yang for their help in our study operation and system development. This work was supported by the National Natural Science Foundation of China (62107007) and Sichuan Science and Technology Program (23NS-FSC4303), China and Beijing Natural Science Foundation (J230013).

REFERENCES

[1] Josephine Barbaro and Cheryl Dissanayake. 2013. Early markers of autism spectrum disorders in infants and toddlers prospectively identified in the Social Attention and Communication Study. *Autism* 17, 1 (2013), 64–86.

[2] Kellen Briot, Adrien Pizano, Manuel Bouvard, and Anouck Amestoy. 2021. New technologies as promising tools for assessing facial emotion expressions impairments in ASD: a systematic review. *Frontiers in psychiatry* 12 (2021), 634756.

[3] Kimberly LH Carpenter, Jordan Hahemi, Kathleen Campbell, Steven J Lippmann, Jeffrey P Baker, Helen L Egger, Steven Espinosa, Saritha Vermeer, Guillermo Sapiro, and Geraldine Dawson. 2021. Digital behavioral phenotyping detects atypical pattern of facial expression in toddlers with autism. *Autism Research* 14, 3 (2021), 488–499.

[4] Keshi Dai. 2007. Early diagnosis of autism through analysis of pre-speech vocalizations. *SIGACCESS Access. Comput.* 89 (sep 2007), 42–46. https://doi.org/10.1145/1328567.1328575

[5] Ryan Anthony Jalova de Belen, Tomasz Bednarz, and Arcot Sowmya. 2021. EyeXplain Autism: Interactive System for Eye Tracking Data Analysis and Deep Neural Network Interpretation for Autism Spectrum Disorder Diagnosis. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) *(CHI EA '21)*. Association for Computing Machinery, New York, NY, USA, Article 364, 7 pages. https://doi.org/10.1145/3411763.3451784

[6] Huiyu Duan, Guangtao Zhai, Xiongkuo Min, Zhaohui Che, Yi Fang, Xiaokang Yang, Jesús Gutiérrez, and Patrick Le Callet. 2019. A Dataset of Eye Movements for the Children with Autism Spectrum Disorder. In *Proceedings of the 10th ACM Multimedia Systems Conference* (Amherst, Massachusetts) *(MMSys '19)*. Association for Computing Machinery, New York, NY, USA, 255–260. https://doi.org/10.1145/3304109.3325818

[7] Centers for Disease Control, Prevention, et al. 2023. Autism prevalence higher, according to data from 11 ADDM communities.

[8] Frances Page Glascoe. 2003. Parents' evaluation of developmental status: how well do parents' concerns identify children with behavioral and emotional problems? *Clinical pediatrics* 42, 2 (2003), 133–138.

[9] Golijeh Golarai, Kalanit Grill-Spector, and Allan L Reiss. 2006. Autism and the development of face processing. *Clinical neuroscience research* 6, 3-4 (2006), 145–160.

[10] Antonio Gulli and Sujit Pal. 2017. *Deep learning with Keras.* Packt Publishing Ltd.

[11] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. 2012. Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems* 25 (2012).

[12] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. 1998. Gradient-based learning applied to document recognition. *Proc. IEEE* 86, 11 (1998), 2278–2324.

[13] Shuo Liu, Yue Wang, and Yongning Song. 2023. Atypical facial mimicry for basic emotions in children with autism spectrum disorder. *Autism Research* (2023).

[14] Wenbo Liu, Ming Li, and Li Yi. 2016. Identifying children with autism spectrum disorder based on their face processing abnormality: A machine learning framework. *Autism Research* 9, 8 (2016), 888–898.

[15] Catherine Lord, Mayada Elsabbagh, Gillian Baird, and Jeremy Veenstra-Vanderweele. 2018. Autism spectrum disorder. *The lancet* 392, 10146 (2018), 508–520.

[16] Angelina Lu and Marek Perkowski. 2021. Deep Learning Approach for Screening Autism Spectrum Disorder in Children with Facial Images and Analysis of Ethnoracial Factors in Model Development and Application. *Brain Sciences* 11, 11 (2021). https://doi.org/10.3390/brainsci11111446

[17] Rhiannon Luyster. 2012. Autism Diagnostic Observation Schedule, (ADOS-2) Manual (Part II): Toddler Module. (2012).

[18] Natasha Malik-Soni, Andrew Shaker, Helen Luck, Anne E Mullin, Ryan E Wiley, ME Suzanne Lewis, Joaquin Fuentes, and Thomas W Frazier. 2022. Tackling healthcare access barriers for individuals with autism from diagnosis to adulthood. *Pediatric Research* 91, 5 (2022), 1028–1035.

[19] Hiromi Nemoto, Pavel Korshunov, Philippe Hanhart, and Touradj Ebrahimi. 2015. Visual attention in LDR and HDR images. In *9th International Workshop on Video Processing and Quality Metrics for Consumer Electronics (VPQM)*.

[20] Chiugo Okoye, Chidi M Obialo-Ibeawuchi, Omobolanle A Obajeun, Sarosh Sarwar, Christine Tawfik, Madeeha Subhan Waleed, Asad Ullah Wasim, Iman Mohamoud, Adebola Y Afolayan, and Rheiner N Mbaezue. 2023. Early diagnosis of autism spectrum disorder: A review and analysis of the risks and benefits. *Cureus* 15, 8 (2023).

[21] Diana L Robins, Deborah Fein, and Marianne Barton. 2009. Modified checklist for autism in toddlers, revised, with follow-up (M-CHAT-R/F) TM. *LineageN* (2009).

[22] Noah J Sasson and Emily W Touchstone. 2014. Visual attention to competing social and object images by preschool children with autism spectrum disorder. *Journal of autism and developmental disorders* 44 (2014), 584–592.

[23] Eric Schopler, Mary E Van Bourgondien, Glenna Janette Wellman, and Steven R Love. 2010. Childhood Autism Rating Scale–Second Edition (CARS-2). *Los Angeles, CA: Western Psychological Services* (2010).

[24] Da-Yea Song, So Yoon Kim, Guiyoung Bong, Jong Myeong Kim, and Hee Jeong Yoo. 2019. The use of artificial intelligence in screening and diagnosis of autism spectrum disorder: a literature review. *Journal of the Korean Academy of Child and Adolescent Psychiatry* 30, 4 (2019), 145.

[25] Mark L Sundberg. 2008. *VB-MAPP Verbal Behavior Milestones Assessment and Placement Program: a language and social skills assessment program for children with autism or other developmental disabilities: guide.* Mark Sundberg.

[26] Uzma Haque Syeda, Ziaul Zafar, Zishan Zahidul Islam, Syed Mahir Tazwar, Miftahul Jannat Rasna, Koichi Kise, and Md. Atiqur Rahman Ahad. 2017. Visual Face Scanning and Emotion Perception Analysis between Autistic and Typically Developing Children. In *Proceedings of the 2017 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers* (Maui, Hawaii) *(UbiComp '17)*. Association for Computing Machinery, New York, NY, USA, 844–853. https://doi.org/10.1145/3123024.3125618

[27] Samy Tafasca, Anshul Gupta, Nada Kojovic, Mirko Gelsomini, Thomas Maillart, Michela Papandrea, Marie Schaer, and Jean-Marc Odobez. 2023. The AI4Autism Project: A Multimodal and Interdisciplinary Approach to Autism Diagnosis and Stratification. In *Companion Publication of the 25th International Conference on Multimodal Interaction*. 414–425.

[28] Diana Weiting Tan, Murray T Maybery, Syed Zulqarnain Gilani, Gail A Alvares, Ajmal Mian, David Suter, and Andrew JO Whitehouse. 2020. A broad autism phenotype expressed in facial morphology. *Translational psychiatry* 10, 1 (2020), 7.

[29] Saime Tek and Rebecca J Landa. 2012. Differences in autism symptoms between minority and non-minority toddlers. *Journal of autism and developmental disorders* 42 (2012), 1967–1973.

[30] Haishuai Wang, Lianhua Chi, Chanfei Su, and Ziping Zhao. 2022. ASDFace: Face-Based Autism Diagnosis via Heterogeneous Domain Adaptation. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management* (Atlanta, GA, USA) *(CIKM '22)*. Association for Computing Machinery, New York, NY, USA, 4999–5003. https://doi.org/10.1145/3511808.3557170

Table A1: Training results for each epoch (Adam+Preprocessing)

Epoch	Training Loss	Training accuracy	Verification Loss	Verification accuracy
Epoch 1	27.28	0.55	42.54	0.50
Epoch 2	29.54	0.57	29.63	0.57
Epoch 3	28.66	0.58	28.63	0.58
Epoch 4	27.02	0.62	27.05	0.57
Epoch 5	25.36	0.65	25.30	0.58
Epoch 6	24.87	0.66	24.81	0.52
Epoch 7	23.44	0.67	23.32	0.62
Epoch 8	23.09	0.69	23.09	0.65
Epoch 9	22.37	0.72	22.22	0.70
Epoch 10	20.98	0.75	20.93	0.71
Epoch 11	19.24	0.77	19.13	0.74
Epoch 12	18.06	0.75	17.91	0.72
Epoch 13	17.43	0.81	17.25	0.63
Epoch 14	16.87	0.82	16.66	0.72
Epoch 15	16.25	0.85	16.14	0.74
Epoch 16	15.22	0.85	15.10	0.80
Epoch 17	14.88	0.86	14.32	0.86
Epoch 18	14.77	0.88	14.12	0.75
Epoch 19	14.52	0.89	14.01	0.85
Epoch 20	13.89	0.90	13.68	0.88
Epoch 21	13.66	0.88	13.58	0.74
Epoch 22	13.53	0.87	13.34	0.82
Epoch 23	12.41	0.90	12.25	0.92
Epoch 24	11.87	0.91	11.71	0.84
Epoch 25	10.98	0.90	10.65	0.94
Epoch 26	10.64	0.88	10.33	0.73
Epoch 27	10.22	0.90	10.12	0.90
Epoch 28	10.07	0.90	9.98	0.91
Epoch 29	10.03	0.92	10.56	0.62
Epoch 30	10.01	0.90	9.87	0.90
Epoch 31	9.93	0.91	9.81	0.81
Epoch 32	9.57	0.92	9.37	0.83
Epoch 33	9.15	0.91	8.95	0.75
Epoch 34	8.74	0.92	8.66	0.86
Epoch 35	8.24	0.89	8.15	0.80
Epoch 36	7.86	0.90	7.59	0.88
Epoch 37	7.64	0.88	7.47	0.88
Epoch 38	7.32	0.91	7.22	0.84
Epoch 39	6.99	0.90	6.95	0.87
Epoch 40	6.89	0.92	7.13	0.90
Epoch 41	6.64	0.91	6.61	0.90
Epoch 42	6.72	0.90	6.76	0.82
Epoch 43	6.51	0.83	6.49	0.69
Epoch 44	6.33	0.86	6.30	0.84
Epoch 45	6.13	0.85	6.11	0.91
Epoch 46	5.88	0.89	5.81	0.88
Epoch 47	5.68	0.91	5.67	0.92
Epoch 48	5.43	0.90	5.41	0.94
Epoch 49	5.28	0.92	5.26	0.91
Epoch 50	5.11	0.92	5.10	0.93